2021-2022 Fall Term

Highway Engineering Group 5 Problem Session-1

Question 1: A truck is moving along a 8% upgraded road section with a constant speed of 40 km/h. The motor power of the truck is 170kW and the weight of the truck is 5 tons.

a) Find the traction force Ztr.
b) Find the amount of useful load that the truck can carry.
c) If this truck stops on the road, can it move again with the useful load calculated in part b.

Given: rolling resistance coefficient, μ_r=0,030; air resistance coefficient, K=0,070; F=0,9.b.h; b=2,30 m, h=2,70 m; $W_h = 0{,}75.K.F.V^2$; motor efficiency coefficient, η=0,85, Sliding friction coefficient, *f*=0,60; The adherence weight, Q_m=2/3.Q.

Solution:

a) $Ztr = 3600 * \left(\frac{Nm}{V}\right) * \eta$ ➔➔ $Ztr = 3600 * \left(\frac{170}{40}\right) * 0{,}85$ ➔➔ $Ztr = 13005\,N$

b) $Ztr = \sum W$; $\sum W = Wy + Wh + We$

$Wy = \mu_r * Q$ ➔➔ $0{,}030 * Q$ ➔➔ $0{,}03Q$

$W_h = 0{,}75.K.F.V^2$ ➔➔ $0{,}75*0{,}07*(0{,}9*2{,}30*2{,}70)*40^2$ ➔➔ 469,48 N

$We = \left(\frac{Q*s}{100}\right)$ ➔➔ $We = \left(\frac{Q*8}{100}\right)$ ➔➔ $0{,}08Q$

$\sum W = 0{,}03Q + 0{,}08Q + 469{,}48 = 13005$ ➔➔ Q = 113959,27 N ➔➔ 11395,93 kg

Useful load = 11395,93 Kg – 5000 Kg = 6395,93 Kg

c)

According to the two conditions of motion, for a stopping vehicle the following situations can be written:

1. $Z_{tr} < \sum W < Q_m.f \rightarrow$ The vehicle does not move (no motion)

2. $\sum W < Z_{tr} < Q_m.f \rightarrow$ The vehicle starts to move with the normal motion (rolling condition)

3. $\sum W < Q_m.f < Z_{tr} \rightarrow$ The vehicle starts to move while spinning (forward motion with spinning)

4. $Q_m.f < \sum W < Z_{tr} \rightarrow$ The vehicle spins and does not move (spinning without motion)

$Q_{m*} f > Z_{tr}$ (required for starting the motion)

$Q_{m*} f = 2/3 * 113959,27 * 0,6 = 45583,71 N$

$Z_{tr} = \sum W = Wy + Wh + We$

Wh=0 vehicle is not moving

Wy = μ_r * Q ➔➔ 0,030 * 113959,27 ➔➔ 3418,78 N

We=0,08 * Q ➔➔ 0,08 * 113959,27 ➔➔ 9116,74 N

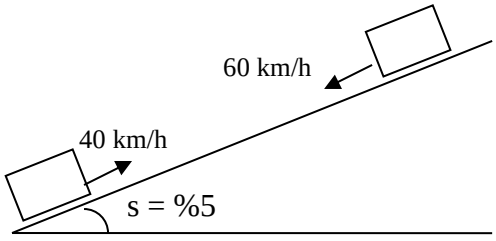
$\sum$W =12535,52 N

45583,71 N > 12535,52 N so it can start moving again.

Question 2: A vehicle is moving along a 5% downgraded one-way road section with a constant speed of 60km/h. At the same time another vehicle has entered this road section in the opposite direction by mistake with a constant speed of 40 km/h. What should be the minimum sight distance in order to prevent the collision ?

Given: Sliding friction coefficient, f = 0.60; Perception-reaction time (sec) t_r = 1 sec (for both vehicles)

Solution:



s = -%5
V_1 = 60 km/h
V_2 = 40 km/h

brake safety distance=l_{fe}

$l_{fe} = 0,278 * V * t_r + 0,00394 * \frac{V^2}{f \mp s}$

$$l_{fe2} = 0.278 * 40 * 1 + 0.00394 * \frac{40^2}{0.6 + 0.05} = 20.82m$$

$$l_{fe1} = 0.278 * 60 * 1 + 0.00394 * \frac{60^2}{0.6 - 0.05} = 42.47m$$

$$l_{\min} = l_{fe1} + l_{fe2} = 20.82 + 42.47 = 63.29m$$

Question 3: A car driver is moving on a slopeless road section with a constant speed of 90 km/h in a rainy day. The driver sees a pedestrian is crossing the road approximately 100 meters ahead.

a) Check if the car hits the pedestrian.

b) If the car hit the pedestrian, what should have been the maximum speed of the car in order to prevent collision ?

Given: Sliding friction coefficient, f = 0.35; Perception-reaction time (sec) t_r = 0,75 s

Solution:

a)

$$l_{fe} = 0{,}278 * V * t_r + 0{,}00394 \frac{V^2}{f \mp s} \rightarrow\rightarrow l_{fe} = 0{,}278 * 90 * 0{,}75 + 0{,}00394 \frac{90^2}{0{,}35 \mp 0}$$

➔➔ 109,95 m hits the pedestrian !!!

b)

In order to prevent collision it should stop within 100 meters.

l_{fe} <= 100 m

$$100 = 0{,}278 * V * t_r + 0{,}00394 \frac{V^2}{f \mp s} \quad \rightarrow\rightarrow 100 = 0{,}278 * V * 0{,}75 + 0{,}00394 \frac{V^2}{0{,}35}$$

➔➔ $0{,}01126 * V^2 + 0{,}2085 * V - 100 = 0$

Solve the equation and max speed;

V_1= 85,50 km/h

Note = $ax^2 + bx + c$ roots of the equation➔➔ $X_{1,2} = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$

Question 4: A car moving on a slopeless road with a constant speed of 90 km/h. This car wants to pass the truck moving in front of the car with a constant speed of 60 km/h.

a) What should be the required distance for the car to pass the truck safely? And what is the total time period for passing ? ($d_i = 8 + 0.3.V_i$)

b) In case of a car coming from the opposite direction with a constant speed of 70 km/h what is the required passing-sight distance?

c) In case of both of the vehicles are moving a speed of 60 km/h and the car in the back passing the front vehicle by accelerating from 60 km/h to 90km/h in 6 seconds. What should be passing time and distance ? Perception-reaction time (sec) t_r = 1 sec

d) In case of both of the vehicles are moving a speed of 60 km/h and the vehicle in the back passing the front vehicle by accelerating from 60 km/h to 90km/h in 6 seconds. And a vehicle is coming from the opposite direction with a constant speed of 70 km/h what is the required passing-sight distance?

Solution:

a)

$d_1=8+0,3.V_1$ ➔➔ 8 + 0,3*90 = 35 m ; $d_2=8+0,3.V_2$ ➔➔ 8 + 0,3*60 = 26 m

$$ts = \frac{d_1 + d_2}{V_1 - V_2} \quad \text{➔➔} \quad ts = \frac{35 + 26}{(90 - 60) * \frac{1}{3,6}} \quad \text{➔➔} \quad \text{ts = 7,32 sn}$$

$ls=ts*V_1$ ➔➔ $ls = \frac{d_1 + d_2}{V_1 - V_2} * V_1$ ➔➔ $ls = \frac{35 + 26}{(90 - 60)} * 90$ ➔➔ ls = 183 m

b)

$ls=ts\ (V_1+V_3)$ ➔➔ $ls = \frac{d_1 + d_2}{V_1 - V_2} * (V_1 + V_3)$ ➔➔ $\frac{35 + 26}{90 - 60} * (90 + 70)$ ➔➔325,33 m

c)

$d_1= d_2 = 26$ m

$t_s=t_r+[2.(d/a)^{0,5}]$ ➔➔ $t_s=1+[2.(26/1,39)^{0,5}]$ ➔➔ 9,65 sn

$$a = \frac{\Delta V}{\Delta t} \quad \text{➔➔} \quad \gamma = \frac{(90 - 60)}{6} * \frac{1}{3,6} \quad \text{➔➔} 1,39\ \text{m/sn}^2$$

$ls=ts*V_2 + 2d$ ➔➔ 9,65*(60/3,6) + 2*26 ➔➔ 212,83 m

d)

$Ls=l_{si} + l_{sk}$ ➔➔ $Ls= ts*V_2 + 2d + V_3*ts$

Ls=9,65*(60/3,6) + 2*26 + (70/3,6)*9,65 ➔➔ Ls= 400,47 m

Question 5: A car moving on a slopeless road with a constant speed of 90 km/h. This car passed the bus moving in front of the car with a constant speed of 70 km/h. Just after the passing car driver noticed an obstacle on the road and pushed the break pedal. The car stopped 1 m before the obstacle.

a) At the beginning of the passing, what is the distance between the car and the obstacle?
b) Check if the bus hits the car during this period?

Given: $d_i=7+0,2.V_i$; Sliding friction coefficient, f = 0.40; Perception-reaction time (sec) t_r = 2 sec (for both vehicles)

Solution: a)

$d_1=7+0,2.V_1$ ➔➔ 7 + 0,2*90 ➔➔ 25 m

$d_2=7+0,2.V_2$ ➔➔ 7 + 0,2*70 ➔➔ 21 m

$ls = \frac{d_1 + d_2}{V_1 - V_2} * V_1$ ➔➔ $ls = \frac{25 + 21}{90 - 70} * 90$ ➔➔ ls = 207 m

$l_{fe} = 0{,}278 * V_1 * t_r + 0{,}00394 \frac{V_1^2}{f \mp s}$ ➔➔ $l_{fe} = 0{,}278 * 90 * 2 + 0{,}00394 \frac{90^2}{0{,}4}$ ➔➔ 129,83m

$l = l_s + l_{fe} + 1$ ➔➔ 207 + 129,83 + 1 ➔➔ 337,83 m

b) $l_{fe} = (0{,}278 * 70 * 2) + (0{,}278 * 70 * 2) + 0{,}00394 \frac{70^2}{0{,}4}$ ➔➔ 126,11 m

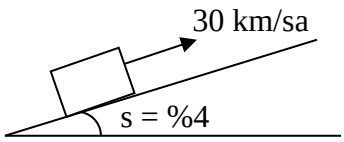
l_f = 129,83 m (automobile)

126,11 < 129,83+21 Therefore, no crash occurs.

Question 6: A loaded truck with a weight of 12 tons, is moving on a 4% upgraded road section with a constant speed of 30 km/h . What should be its motor power (kW) in order to keep moving with constant speed?

Given: rolling resistance coefficient μ_r = 0.025; air resistance coefficient, K = 0.060; F = 0.9.b.h; b = 2.20 m , h = 2.50 m; $W_h = 0.75.K.F.V^2$; motor efficiency coefficient, η = 0.85.

Solution:



m = 12 ton
N_m = ?

The required condition for moving with constant speed $Z_{tr} = \Sigma W$
$\Sigma W = W_y + W_h + W_e$

We : Grade resistance
Wy : rolling resistance
Wh : Air resistance
Q : Weight of the vehicle

$W_y = \mu_r * Q = \mu_r * m * g = 0.025 * 12000 * 10 = 3000$ N

$W_h = 0.75.K.F.V^2 = 0.75 * K * (0.9 * b * h) * V^2$
$W_h = 0.75 * 0.060 * (0.9 * 2.20 * 2.50) * 30^2 = 200.48$ N

$W_e = Q * s = m * g * s = 12000 * 10 * 0.04 = 4800$ N

$\Sigma W = W_y + W_h + W_e = 3000 + 200.48 + 4800 = 8000.48$ N

$$N_m = Z_{tr} . \frac{V}{3600.\eta}$$

$$N_m = \frac{8000.48 * 30}{3600 * 0.85} = 78.44 kW$$

Question 7: A vehicle with a weight of 1050 kg has a Motor Power value of 70 Hp. The dimensions of this vehicle are 1,50 m width and 1,70 m height. Calculate the upgrade level that this vehicle can move with a constant speed of 120 km/h?

Given: rolling resistance coefficient = 0,020 N/N
air resistance coefficient= 0,025
motor efficiency coefficient = 0,75

Solution:

Q = 1050 kg = 10500 N

Horsepower to Kilowatts ➔ N_m = 70 BB = 70 * 0.735 = 51.45 kW

$Z_{tr} = 3600 * \frac{N_m}{V} * \eta = 3600 * \frac{51.45}{120} * 0.75 = 1157.625$ N

$\sum W = W_y + W_h + W_e$

$W_y = \mu_r * Q = 0.020 * 10500 = 210$ N

$W_h = 0.75 * K * F * V^2 = 0.75 * 0.025 * (0.9 * 1.50 * 1.70) * 120^2 = 619.65$ N

$W_e = \frac{Q * s}{100}$ $\qquad Z_{tr} = \sum W$

$1157.625 = 210 + 619.65 + W_e$

$W_e = 327.975$ N

$W_e = \frac{10500.s}{100} = 327.975$ => s = % 3.12

Question 8: A vehicle is moving with a constant speed of 70 km/h along a road section with a sliding friction coefficient value of 0,50. Calculate the brake safety distances for the cases that the road section is 5% downgraded and 5% upgraded. (Perception-reaction time (sec) t_r = 0,75 sn)

Solution:

$l_{fe} = l_r + l_f$

$l_f = 0{,}00394 * \frac{V^2}{f \mp s}$

$l_r = 0{,}278 * V * t_r$

$l_{fe} = 0{,}278 * V * t_r + 0{,}00394 * \frac{V^2}{f \mp s}$

If upgraded

$l_{fe} = 0{,}278 * V * t_r + 0{,}00394 * \frac{V^2}{f + s}$

$l_{fe} = 0{,}278 * 70 * 0{,}75 + 0{,}00394 * \frac{70^2}{0{,}50 + 0{,}05} = 14{,}595 + 35{,}10 = 49{,}695$ m.

If downgraded:

$$l_{fe} = 0{,}278 * V * t_r + 0{,}00394 * \frac{V^2}{f - s}$$

$$l_{fe} = 0{,}278 * 70 * 0{,}75 + 0{,}00394 * \frac{70^2}{0{,}50 - 0{,}05} = 14{,}595 + 42{,}90 = 57{,}495 \text{ m.}$$

Question 9: A bus is moving along a straight road section with a constant speed of 50 km/h. A car moving behind the bus wants to pass the bus by accelerating from 50 km/h to 90 km/h in 7 seconds. What is the time period and distance for safe passing? (t_r = 1 s, d = 7 + 0.3V)

Solution:

7 sec (90-50)*(1/3,6)
1 sec a

a = (90-50) /(3,6*7) = 1,59 m/sec^2

d = 7 + 0,3*50 = 22 m $t_s = t_r + [2.(d/a)^{0,5}] = 1 + [2.(22/1,59)^{0,5}] = 8{,}44$ sec

$$l_{si} = V_2 * t_{si} + 2 * d = \frac{50}{3.6} * 8.44 + 2 * 22 = 161.22m$$

2021-2022 Fall Term

Highway Engineering Group 5 Problem Session-2

Question 1: A new two way double laned highway have started to operate in 1998. First year 1.500.000 vehicles, second year 1.590.000 vehicles have used the highway. For each year the heavy vehicle rate is given as 20 %.

a) Check if it achieved to reach 20 years of service life predicton which is calculated in design stage.
b) If the right of way is sufficient, in order to achieve predicted life span, how many lanes should be added in which year?

Given: Necessary arrangments have been performed within the service volume table below. The highway is required to provide C service level at the end of its life span. K=1/6; constant traffic increase; 1 light vehicle = 1 bo, 1 heavy vehicle= 3 bo,

Service Level	Service Volume (bo/h/ln)		
	2 way 2 lanes	3 lanes	2x2 lanes
A	100	200	400
B	200	500	700
C	600	750	1000
D	800	1000	1500
E	1000	1400	1900

Solution:

a) $T_{1998} = \frac{1{,}500{,}000 * (0{,}80 * 1 + 0{,}20 * 3)}{365 * 6}$ =959 bo/h

$T_{1999} = \frac{1{,}590{,}000 * (0{,}80 * 1 + 0{,}20 * 3)}{365 * 6}$ =1017 bo/h

$T_n = T_o * (1+r)^n$ $\quad 1017 = 959 * (1+r)^1$ $\quad$ r = %6

For C- Service level:

$2 * 600 = 959 * (1 + 0{,}06)^n$ $\quad \ln \frac{1200}{959} = n * \ln(1{,}06)$ $\quad$ n=3,85 years

3,85 years < 20 years therefore life span prediction fails.

b) 1 lane should be added after 3,85 years (it will be 3 lanes)

$3 * 750 = 959 * (1 + 0{,}06)^n$ $\quad$ n=14,64 years

So after n=14,64 years one more lane should be added.

In 1998 highway started as 2 lanes

In 1998+3,85 highway had 3 lanes

In 1998+14,64highway became (2x2 lanes)

Question 2: The number of the vehicles passing through a Two-way double lane highway in 1987 and 2002 are in the limits of service levels B and C, respectively.

a) In which year this highway will reach the D service level ?

b) Compare the traffic densities of the complete platform for 1987 and 2002.

c) How long would it be possible to serve at level A If the counted number of vehicles passing throughout the highway would be 300.000 veh/year for 1985.

Given: Necessary arrangments have been performed within the service volume table below. Ignore the ratio of heavy vehicles. Increase ratio of traffic is constant. K=1/6

Service Level	Service Volume (veh/h/ln)	Operational Speed (km/h)
A	100	100
B	200	85
C	550	65
D	800	50

a)

B service level in 1987

C service level in 2002

$T_{2002} = T_{1985} * (1+r)^{15}$ $\quad 550 = 200 * (1+r)^{15}$ $\quad$ r = % 7

For Servicel level D ;

$800 = 200 * (1+0{,}07)^{n}$ $\quad \ln \frac{800}{200} = n * \ln(1{,}07)$ $\quad$ n=20,5 yıl

After 20,5 years the service level becomes D .

b)

q=k*u (Volume=Density*Speed)

$k_{1987} = \frac{200*2}{85}$ = 4,71 == 5 veh/km $\quad k_{2002} = \frac{550*2}{65}$ = 16,92 == 17 veh/km

c)

$T_{1985} = \frac{300{,}000}{365*6}$ = 137 veh/h

Free Flow only possinble in Service level A !!!

For service level A;

$2*100 = 137*(1+0{,}07)^{n}$ $\quad \ln \frac{200}{137} = n * \ln(1{,}07)$ $\quad$ n=5,59 years

It is possible to keep free flow for 5,59 years.

Question 3 : On a rolling terrain, there is a road that consists of double way, two lanes on each side. 2600 vehicles passing in one direction hourly including 18 % truck and 5 % bus type. Lane width is 3,30 m and banquet width is 2 m. Max allowable speed is 80 km/h. Are the number of lanes sufficient to serve at level D ?

Service level Data for multiple lane roads

Service Level	Density (veh/km/ln)	Vp=110 km/h			Vp=100 km/h			Vp=80 km/h		
		$V_{Ort.}$ (km/h)	Q/C	q_m (veh/h/ln)	$V_{Ort.}$ (km/h)	Q/C	q_m (veh/h/ln)	$V_{Ort.}$ (km/sa)	Q/C	q_m (veh/h/ln)
A	≤7,5	≥91	0.36	700	≥80	0.33	650	-	-	-
B	≤12,5	≥85	0.54	1,100	≥77	0.50	1,000	≥67	0.45	850
C	≤18,8	≥80	0.71	1,400	≥70	0.65	1,300	≥62	0.60	1,150
D	≤26,3	≥64	0.87	1,750	≥64	0.80	1,600	≥56	0.76	1,450
E	≤41,9	≥48	1.00	2,000	≥48	1.00	2,000	≥45	1.00	1,900
F	>41,9	<48	d	d	<48	d	d	<45	d	d

d: very inconsistent

Factors used for narrowlane and banquets for multiple lane roads (f_w)

Distance to obstacle (m)*	Factors (f_w)							
	obstacle on one side				obstacle on both sides			
	Lane width (m)							
	3.7	3.3	3.0	2.7	3.7	3.3	3.0	2.7
divided multpiple lane road two lanes each direction								
≥2,0	1.00	0.97	0.91	0.81	1.00	0.97	0.91	0.81
1.0	0.98	0.95	0.89	0.80	0.97	0.94	0.88	0.78
0	0.90	0.87	0.82	0.73	0.81	0.79	0.74	0.66
divided multpiple lane road three lanes each direction								
≥2,0	1.00	0.96	0.89	0.78	1.00	0.96	0.89	0.78
1.0	0.98	0.94	0.88	0.77	0.97	0.93	0.86	0.76
0	0.94	0.91	0.85	0.74	0.91	0.87	0.81	0.70
undivided multpiple lane road two lanes each direction								
≥2,0	1.00	0.95	0.89	0.77	N	N	N	N
1.0	0.97	0.93	0.87	0.76	N	N	N	N
0	0.88	0.85	0.80	0.70	0.81	0.79	0.74	0.66
undivided multpiple lane road three lanes each direction								
≥2,0	1.00	0.95	0.89	0.77	N	N	N	N
1.0	0.98	0.94	0.88	0.76	N	N	N	N
0	0.94	0.90	0.83	0.72	0.91	0.87	0.81	0.70

* Take average if the obstacles have different distances

N: use factors that obstacle on one side

Vehicle Type	Unit automobile values of Bus and Trucks		
	Terrain		
	Straight	Rolling	Mountain
Truck	1.7	4	8
Bus	1.5	3	5

	Factors reflecting environmental features (fe)	
Residantial type	Road type	
	Divided	Undivided
Rural	1	0.95
Suburb	0.9	0.8

Driver Factors (fp)	
Driver Type	(Fp)
Usual Driver	1
Rare Driver	0.75-0.90

Solution:

Traffic in one direction : 2600 veh/h
Terrain : Rolling
Truck rate : %18
Bus rate : %5
Lane width : 3.30 m
Banquet width : 2 m
Driver Feature : usual driver
Project speed : 80 km/h
Service level : D

$Q_{SFi} = C_j \cdot (Q/C)_i \cdot N \cdot f_w \cdot f_{HV} \cdot f_e \cdot f_p$

Q_{SFi} :For service level (i), service volume in one direction (veh/h).
C_j:If the design speed is 80 km/h; C_j = 1900 veh/h/ln; if the design speed is ≥100 km/h; C_j = 2000 veh/h/ln;
(Q/C)i: For service level (i) volume/capacity ratio
N: Number of lane for each direction
f_w :Adjustment factor for lane width and banquet width
f_{HV} :Adjustment factor for heavy vehicle traffic
f_e:Adjustment factor for environmental feature
f_p:Adjustment factor for driver feature

Project speed=80 km/h C_j=1900 veh/h/ln

Q/C = 0.76
f_w = 0.95

$$f_{HV} = \frac{1}{\left[1 + P_T(E_T - 1) + P_B(E_B - 1)\right]} = \frac{1}{\left[1 + 0.18\ (4-1) + 0.05(3-1)\right]} = 0.61$$

f_e = (undivided road) = 0.95
f_p = (usual driver) = 1
$Q_{SFi} = C_j \cdot (Q/C)_i \cdot N \cdot f_w \cdot f_{HV} \cdot f_e \cdot f_p$ put into equation

2600 = 1900. 0.76 . N . 0.95 . 0.61 . 0.95 . 1

N= 3.27 -> According to that a road with 4 lanes required. Add 2 more lanes at one direction.